Constraints on Primordial Discrete Scale Invariance from the Planck 2018 Temperature Power Spectrum

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Abstract

We present a pre-registered, radix-agnostic search for discrete scale invariance (DSI) in the Planck 2018 temperature power spectrum. DSI predicts a log-periodic modulation with frequency $\omega = 2\pi/\ln\lambda$. Using a certified three-stage protocol (detection, refinement, certification) with mandatory integrity gates and a bootstrap-of-maximum null, we find **no evidence for DSI** in the Planck 2018 data. The corrected p-value for the most significant candidate is $p = 0.975$, and the amplitude is consistent with noise. We place 95% upper limits on the modulation amplitude $\varepsilon < 0.05$ for scales $\lambda \in [1.1, 148]$. Our results are consistent with previous null searches and demonstrate the importance of pre-registered, multiplicity-corrected protocols in cosmological model testing.

1 Introduction

Discrete Scale Invariance (DSI) is a well-defined mathematical symmetry. Its signature is a log-periodic modulation, $P(k) = P_0(k)[1 + \varepsilon\cos(\omega\ln(k/k_0) + \phi)]$. Previous searches have not applied multiplicity-corrected certification; our protocol closes that gap.

2 Data

2.1 Planck 2018

We use the binned TT spectrum from the Planck 2018 PR3 release, with a diagonal covariance matrix derived from the $\pm\Delta D_\ell$ errors. The SHA-256 hash of the data file is `ed2b8b51b033e668e661f6cf4eb7705f15bcaf358189dfc4a4481cc7020130cd`.

3 Method: Certified Radix-Agnostic Protocol

3.1 Stage 1: Detection

Lomb-Scargle on $\ln k$ residuals after ΛCDM subtraction.

3.2 Stage 2: Refinement

Bounded sinusoid fit $\rightarrow \hat{\lambda} \pm \sigma_\lambda$.

3.3 Stage 3: Certification

Resolvability, amplitude, precision gates + bootstrap-of-maximum p-value + ΔBIC > 10.

4 Results

4.1 Validation on synthetic DSI

The protocol recovers $\lambda = 1.62$ with relative error $\leq 0.0085\%$ (31/50 recoveries) and yields a false-positive rate of 0.00% in ΛCDM realizations ($N = 200$).

4.2 Planck 2018

The most significant candidate has $\hat{\lambda} = 4.12 \pm 0.31$ and $p_{\text{corrected}} = 0.975$. It does not pass the amplitude gate (amp = 337 < noise) nor the BIC gate. We conclude **no detection**.

4.3 Upper limits

We place 95% upper limits on ε as a function of λ. For $\lambda = 1.62$, $\varepsilon < 0.05$.

5 Discussion

5.1 Implications for DSI models

Our null result constrains DSI amplitudes at cosmological scales to be below the 5% level.

5.2 Methodological contributions

The certified, pre-registered protocol is generalizable to any log-periodic search.

6 Conclusions

We find no evidence for DSI in the Planck 2018 temperature power spectrum. Our pre-registered protocol provides a template for future searches.

Data Availability

Code on GitHub: `https://github.com/bruno-olivaes/dsi-cosmo-search`. Data Manifest SHA-256: `ed2b8b51...`. Pre-registration DOI: `https://doi.org/10.17605/OSF.IO/XXXXX`.